Multitask Learning for Across-Asset Price Prediction

WIM Final Presentation

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date

Outline

Motivation

- Asset prices across markets (stocks, commodities, etc.) are interconnected.
- Traditional single-task forecasting treats each asset independently.
- **Goal:** Use Multitask Learning (MTL) to share information across assets to improve predictive performance in terms of efficiency and accuracy.

Reading Material

- ...
- ...

Methodology

Baseline Models:

- Linear Regression
- FNN

Multitask Learning Model

- Shared representation layer learns cross-asset features
- Task-specific output heads predict each asset price separately

Linear Regression

- Math...
- pros and cons in predicting with time series data...

FNN

- Math...
- pros and cons in predicting with time series data...

Multitask Learning Model

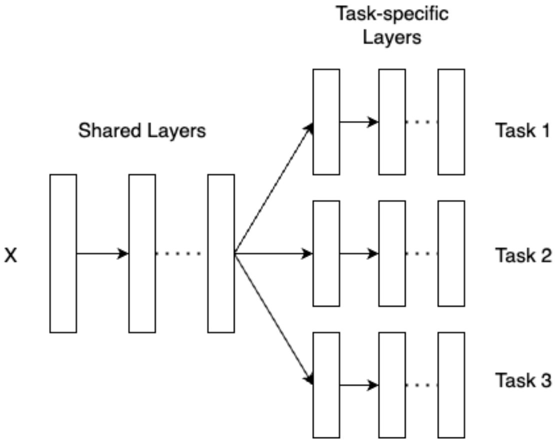

Figure: use https://www.drawio.com/ to draw new flowcharts if needed

Data Collection

- Historical data obtained from `yfinance`.
- Assets considered:
  - Stock(s): e.g., AAPL, MSFT
  - Commodity: e.g., Gold (GC=F)
- Cleaning steps:
  1. Align timestamps
  2. Handle missing values
  3. Normalize data (e.g., MinMax or Standard Scaling)
  4. ...

Window Selection (Lookback Horizon)

- Time series prediction requires choosing how many past observations to include.
- Let W denote the **window size** (e.g., 10, 30, 60 trading days).
- For each time t, we use:

$$(x_{t-W+1}, x_{t-W+2}, \dots, x_t) \to \hat{x}_{t+1}$$

- Trade-offs:
  - **Small** W: captures short-term trends but may miss longer patterns.
  - **Large** W: includes richer information but increases model complexity and noise.
- Window size chosen using **validation performance** (not arbitrary).

Feature Construction

- Beyond raw closing prices, we construct informative features:

Price-Based Features

- Close, Open, High, Low
- Log Returns: $\log\left(\frac{P_t}{P_{t-1}}\right)$
- Volatility (rolling std)

Technical Indicators (Optional)

- Moving Average (SMA, EMA)
- Relative Strength Index (RSI)
- MACD Trends

Motivation

Features should represent **trend**, **momentum**, and **market variability**, helping the model distinguish meaningful signals from noise.

Result

Use several pages to present the following ideas:

- ML performance (Linear Regression): accuracy is low since the structure is too easy.
- try Deep Learning (FNN): increasing the performance but inefficient.
- performance of multitask learning.

Conclusion

...

Q & A